2024.10.30

Vision-Based Intelligent Gait Environment Detection Algorithm for Assisting Stroke Patients with a Soft Ankle Exosuit

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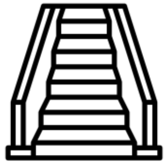
I. Introduction

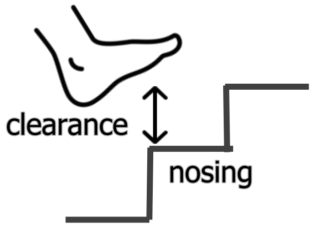
Stroke Patients

- functional impairments & gait disturbances
- foot drop : <u>reduction in foot clearance</u> (swing phase of gait cycle)



stair ambulation

- common activity
- <u>challenging task</u> : joint range, muscular moment ↑
- ascent: <u>insufficient clearance</u> between foot & stair nosing



➡

Lower Limb Exoskeleton

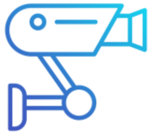
- Assisting gait and rehabilitation
- Crucial for activities of daily living (ADL)

I. Introduction

Exoskeleton

- inertial measurement unit (IMU) & insole sensor
  - ➢ provide real-time gait phase information
  - ➢ Limitation: <u>Lack of prior environmental information</u>
- vision – based perception

human
peripheral vision



exoskeleton
vision system

- ✓ visual recognition of step location
- ✓ continuous monitoring stage
- ✓ awareness of surrounding environment
- ✓ locomotion guidance

I. Introduction

Deep learning-based methodologies

influence robotics field

environment classification + exoskeleton

[challenge]

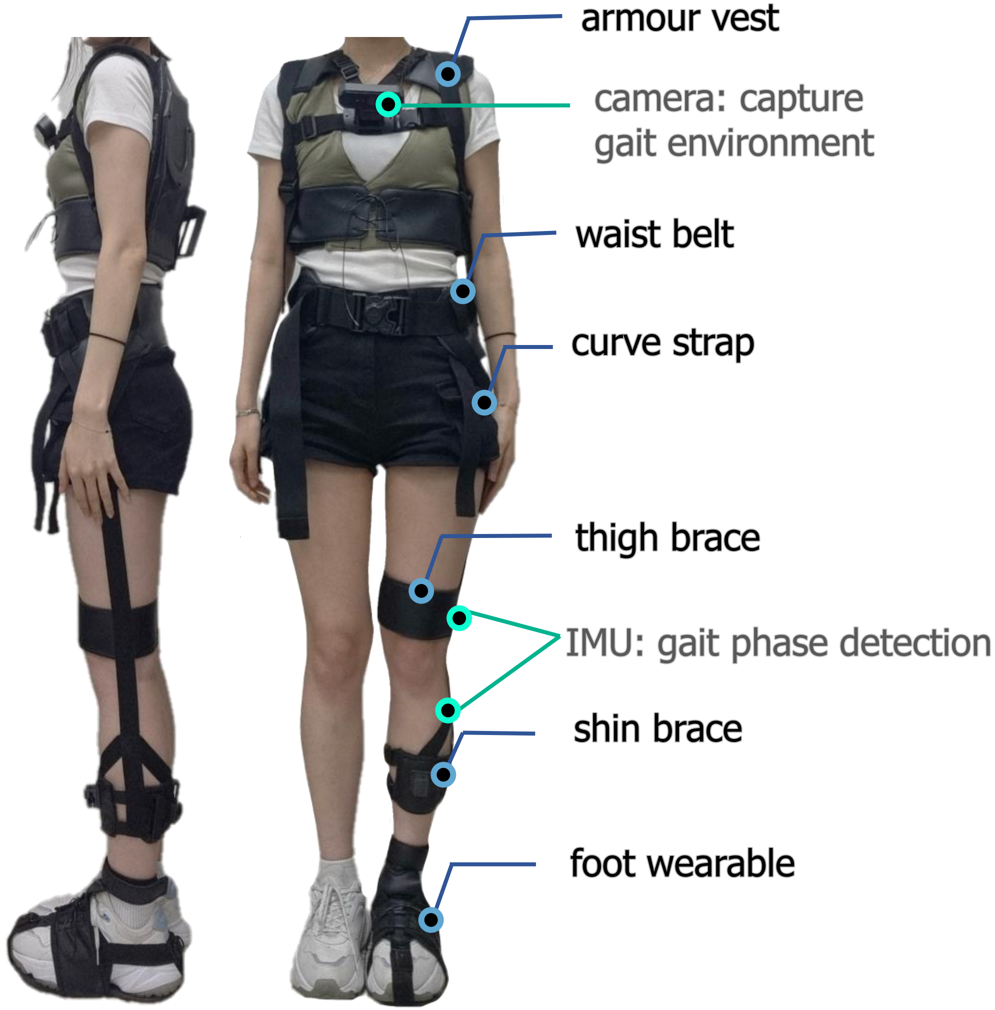
- transition phase accuracy
- stroke patient consideration

Vision-based Intelligent gait environment detection algorithm for ankle exosuit

- Integration of deep learning-based environment classification into exoskeleton application
- detailed gait environment classification (YOLOv8) + spatio-temporal feature extraction (LSTM)
- <u>soft ankle exosuit</u> aimed at <u>stroke patients</u>

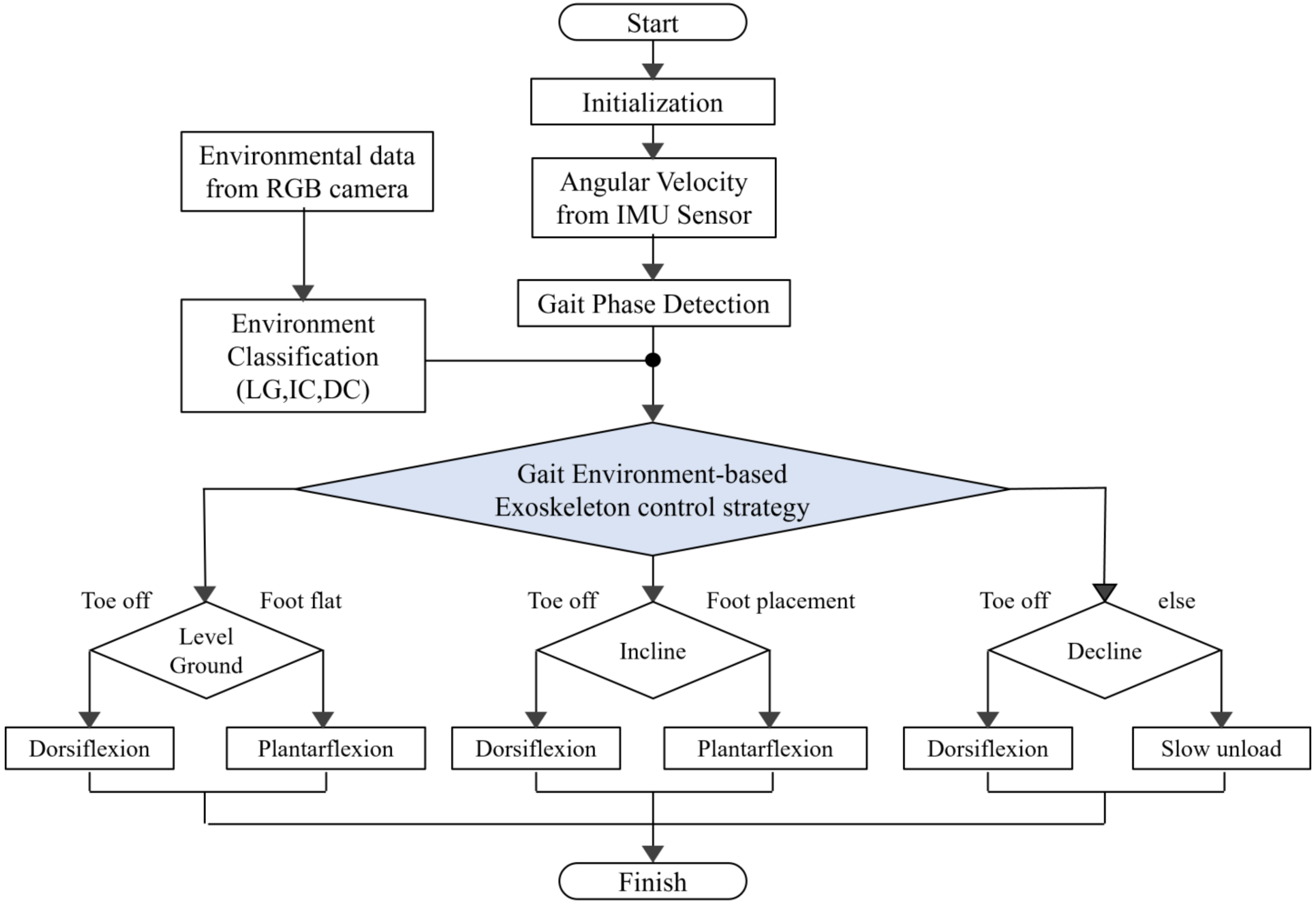
II . Soft Exosuit

Exosuit Design



- flexible and lightweight materials
- Bowden cable-driven method

Control Strategy



flowchart of gait-environment-based exoskeleton control strategy

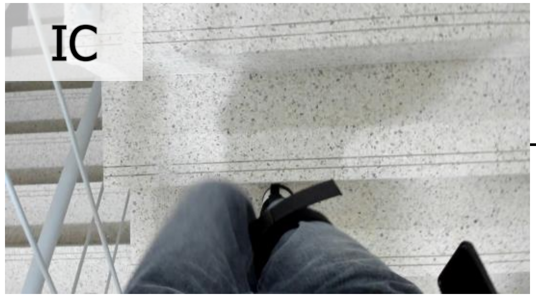
III . Environment Recognition System

Data Acquisition · Class Definition

[customized dataset]

- level ground and stair environment
- 960 x 540 pixel
- training: 16,000 | valid: 2,000 | test: 2000

  -> 8:1:1 train valid test



IC prev	initial incline
IC	mid-incline
IC next	final incline

LG	level ground

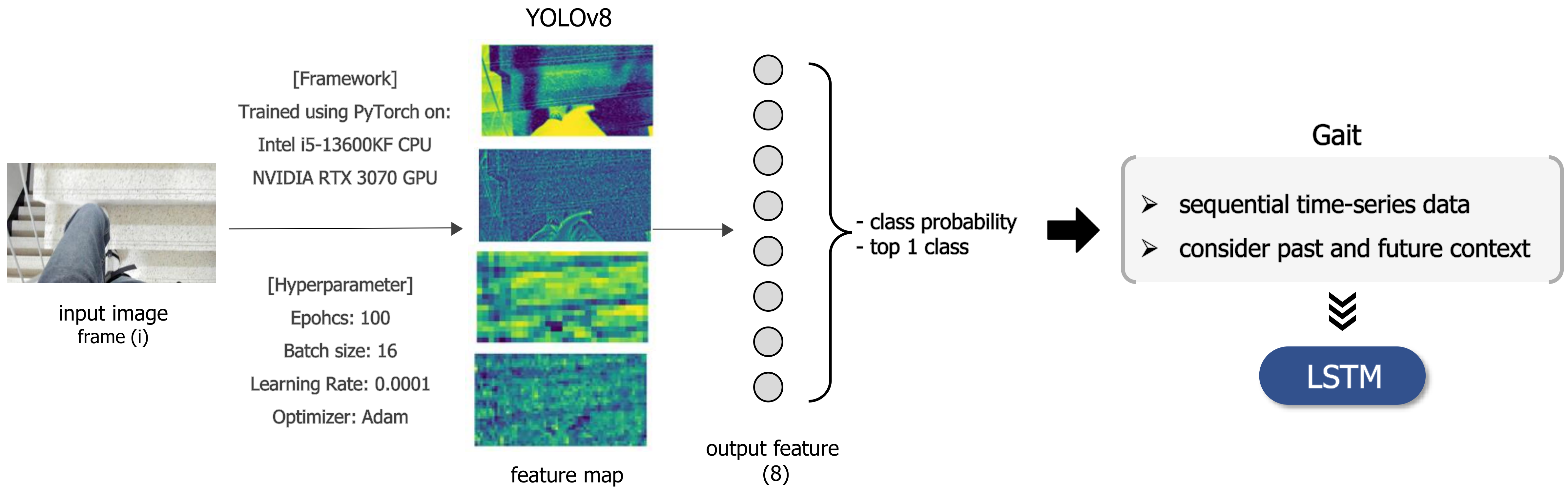
DC prev	initial decline
DC	mid-decline
DC next	final decline

III . Environment Recognition System

YOLOv8 Environment Classification

- object detection framework: one-stage detection approach
- provide <u>real-time contextual insight</u> for frame

Rapid Detection: critical for detecting gait environment transitions
Spatial analysis: Effective in analyzing global spatial information



III . Environment Recognition System

LSTM-Based Spatio-Temporal Feature Extraction

- extract temporal features: adapting to environment transitions

Data Preparation

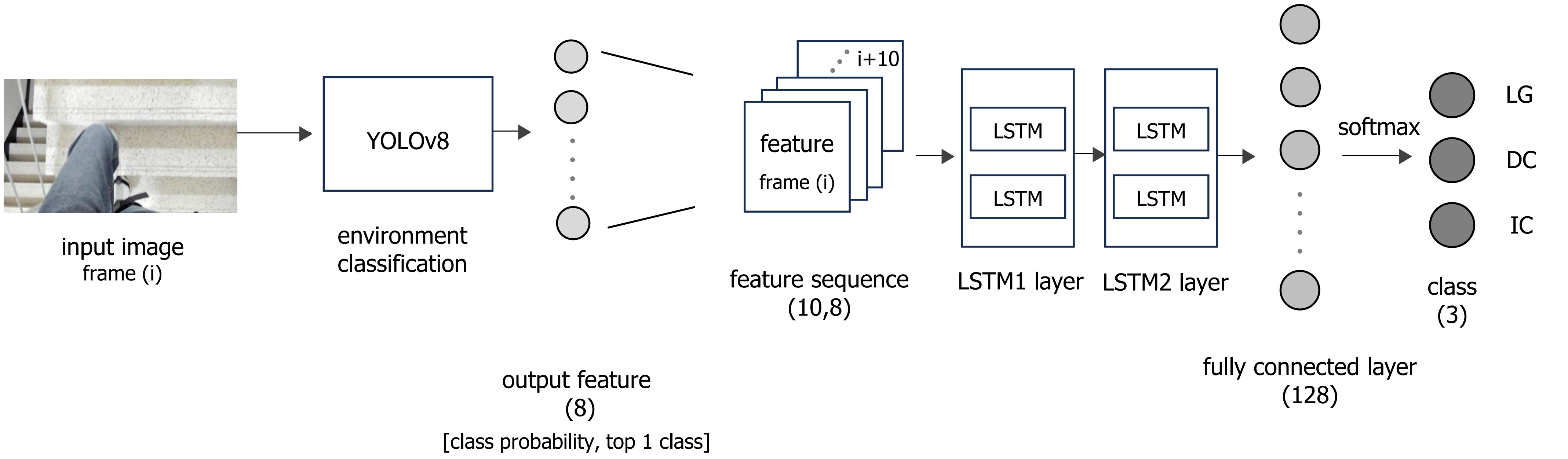
- Frame Sampling: **20 FPS**
- 1 class for 10 frames, with a stride of 5 (**4 labels/sec**)
- Sequence shape: **(1714, 10,8)**

Class Labeling

- **Most probable Class:** Chosen from YOLO frame-wise probabilities
- **Maximum Voting & Average Voting**

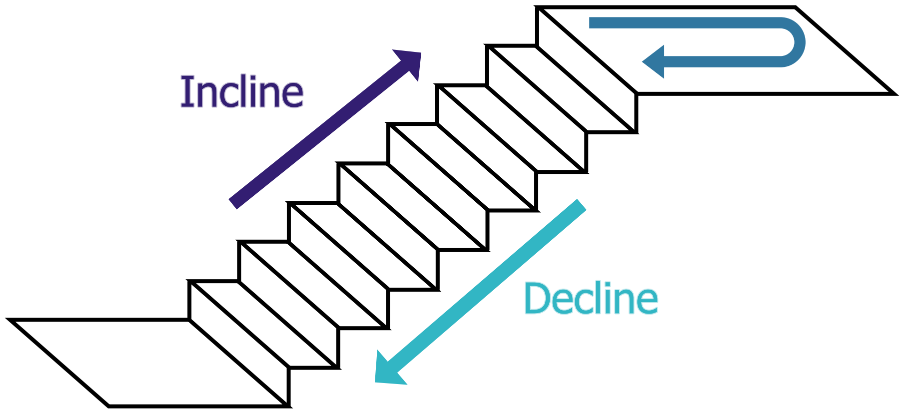
Model Training

- 100 epochs
- batch sizes : 32
- learning rate: 0.001
- Adam optimizer



IV . Experiments

Experimental Setup



Condition

1) without exosuit (*No Exo*)
2) Exosuit worn but unpowered (*Exo Off*)
3) Exosuit worn and powerd (*Exo On*)

10-minute familarization with exosuit

20-minute rest between conditions

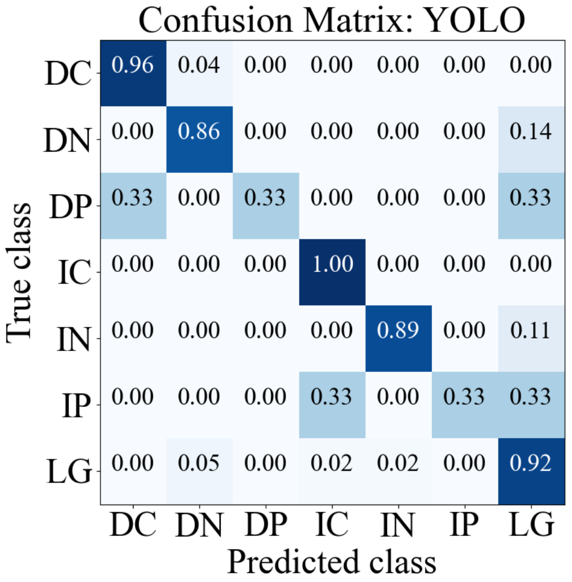
- ➢ gait environment frame

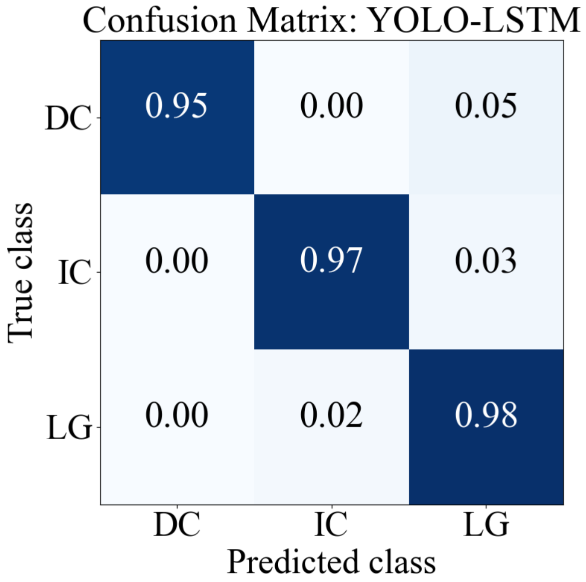
- ➢ Muscle Activation
  - ✓ MVC Normalization
  - ✓ fourth-order Butterworth filter (6Hz cutoff frequency)

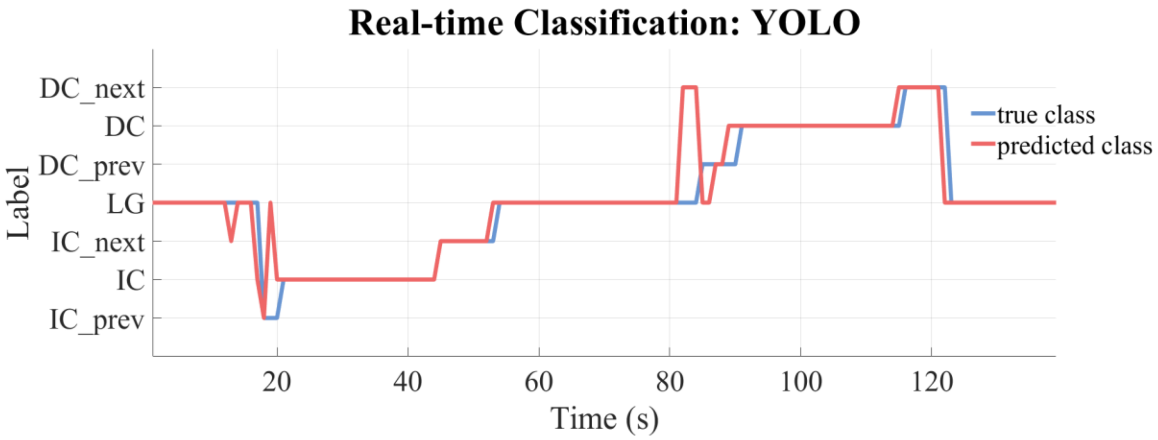
- ➢ ankle joint angles (sagittal plane)

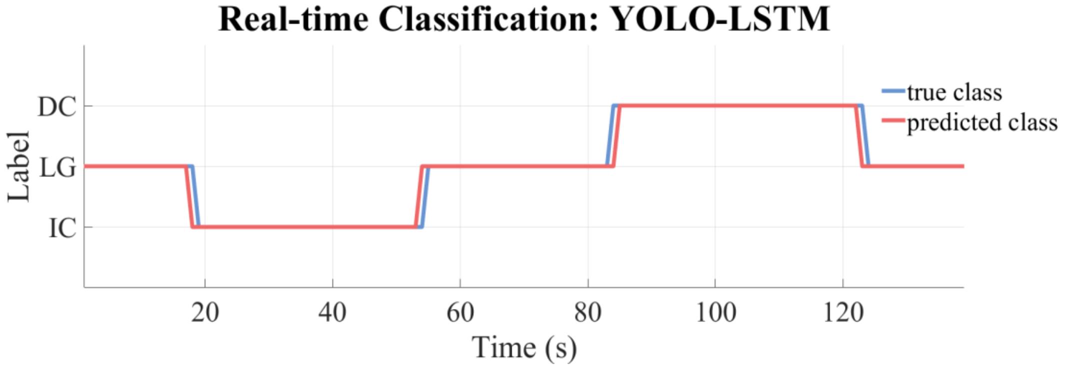
V . Results

Real-Time Environment Classification Performance









YOLO

Low Recall in Transition Classes (IC_{prev} , DC_{prev})

Overall F1 score : **0.774** - transition misclassification

YOLO - LSTM

Overall F1 score : **0.971** - balanced classification performance

YOLO

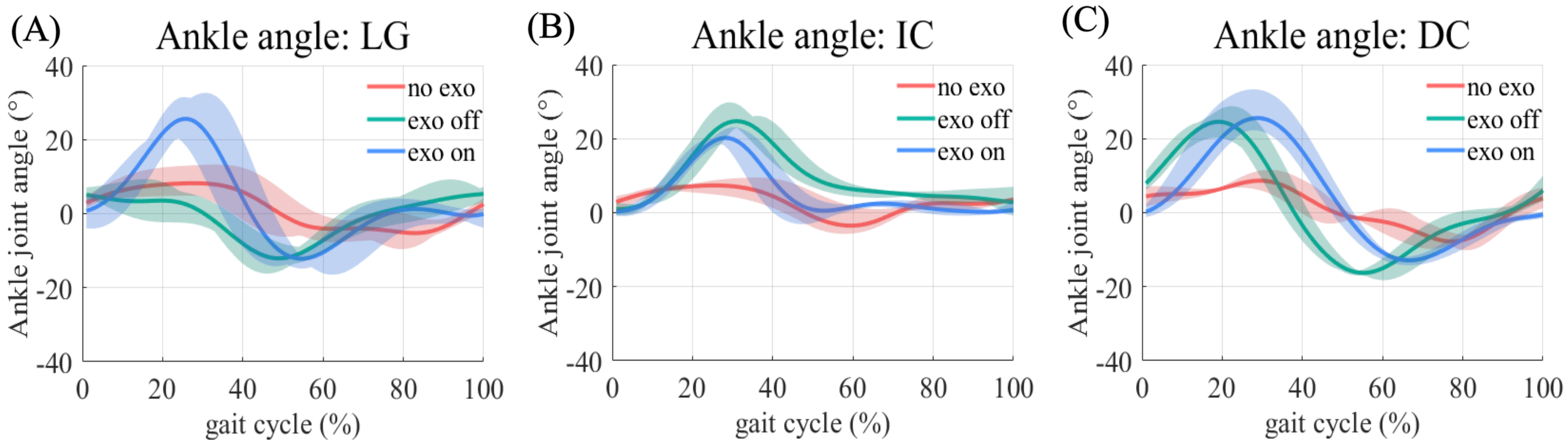
Misclassified LG as stairs 1.5s (6 frames) before stair transition

YOLO - LSTM

rapid and accurate classification during the transition phase

V . Results

Joint Kinematics



peak ankle joint dorsiflexion angle

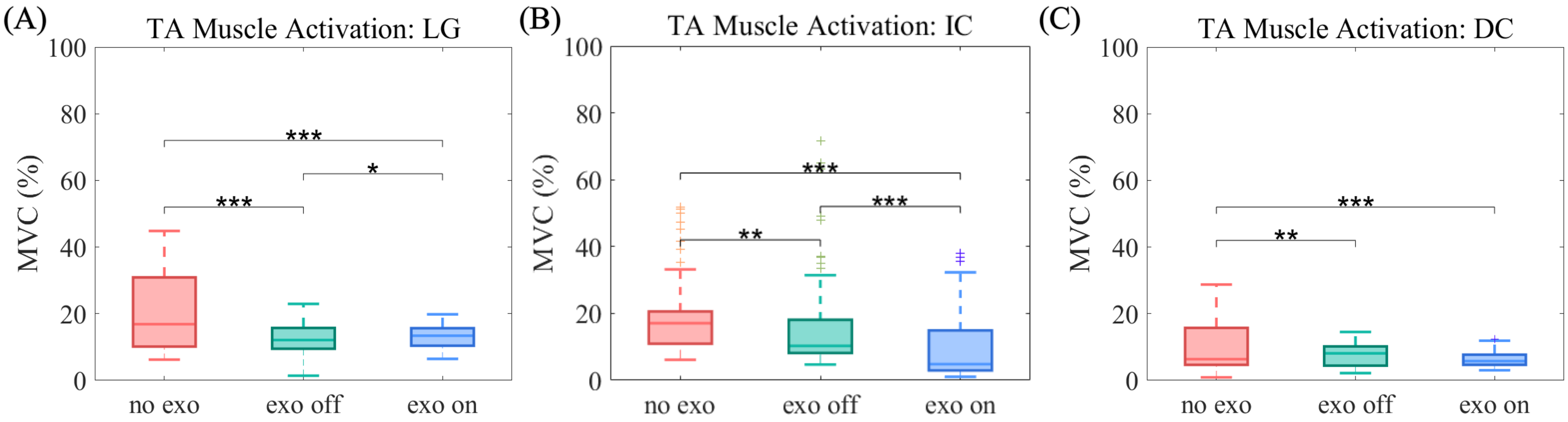
	LG	IC	DC
no exo	8.88° ± 3.95°	7.92° ± 1.48°	9.14° ± 2.27°
exo off	6.74° ± 2.08°	25.99° ± 2.78°	25.08° ± 3.63°
exo on	**27.13° ± 5.71°**	**21.06° ± 2.08°**	**25.65° ± 5.16°**

LG: **significant increase** with the *exo on*

IC: **increase** with the *exo on, exo off*

V . Results

Muscle Reduction



average TA muscle activation

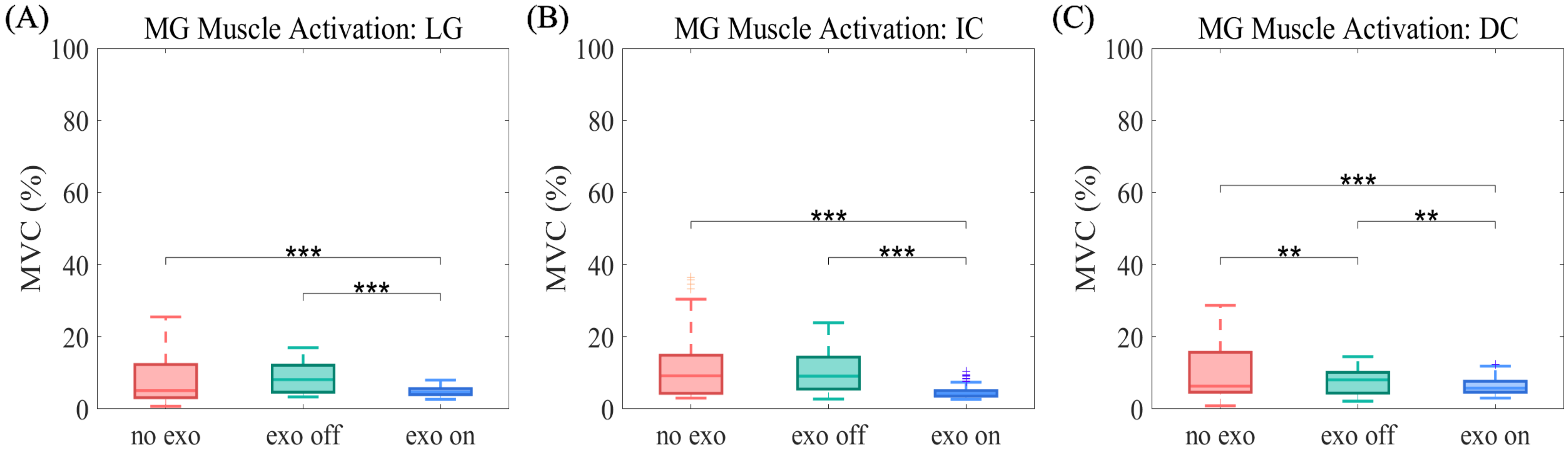
	LG	IC	DC
no exo	20.63 ± 11.56%	18.44% ± 9.92%	10.75% ± 4.25%
exo off	12.77% ± 5.02%	15.24% ± 12.77%	12.34% ± 10.93%
exo on	**13.22% ± 3.43%**	**9.79% ± 10.12%**	**12.09% ± 4.81%**

LG, IC: **reduction** in *exo off* , *exo on* condition

IC: slight increases with exosuit

V . Results

Muscle Reduction



average MG muscle activation

	LG	IC	DC
no exo	7.88% ± 6.61%	11.71% ± 8.83%	9.97% ± 7.35%
exo off	8.51% ± 4.04%	10.66% ± 5.99%	7.62% ± 3.41%
exo on	**4.87% ± 1.21%**	**4.70% ± 1.69%**	**6.41% ± 2.19%**

LG, IC, DC: **reduction** in *exo on* condition

V . Conclusion

Conclusion

- vision-based detection algorithm for an ankle exosuit, focusing on improving gait support for stroke patients
- integration of YOLOv8 for environment classification and LSTM for spatio-temporal recognition
- rapid and accurate classification during gait environment transition
- increased ankle dorsiflexion / Reduced TA and MG muscle activation

Future work

- Validate across diverse real-world scenarios and multiple stroke patients
- Enhance the effectiveness and generalization ability of the exosuit by improving environment classification algorithm

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Thank You

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